

MAT41-1

Linear Algebra IV

The Singular Value Decomposition and its applications

Every matrix, no exceptions, is a rotation, then a stretch along axes, then another rotation — and almost all of data science is reading that sentence carefully.

Albert School — 30 hours

Preface

Across three courses you have been taking matrices apart. In **Linear Algebra I** (MAT11-3) a matrix was a linear map, with a column space and a null space and a rank tying them together. In **Linear Algebra II** (MAT21-1) you learned to look **inside** a square matrix with eigenvalues and eigenvectors, and to write $A = PDP^{-1}$ whenever enough independent eigenvectors existed — exposing the directions a map merely scales. In **Linear Algebra III** (MAT31-1) you added a scalar product, and with it length, angle, and orthogonality; the **spectral theorem** then promised that every real **symmetric** matrix is not just diagonalizable but **orthogonally** diagonalizable, $A = PDP^T$ with P orthogonal and the eigenvalues real.

Eigendecomposition is powerful but fragile. It needs the matrix to be **square**. It can fail to exist (too few eigenvectors). Its P is generally not orthogonal, so it mixes the clean geometry of MAT31-1 with the messier algebra of MAT21-1. And the data matrices of the real world — a spreadsheet of m customers by n products, an image of m rows by n columns of pixels — are almost never square and almost never symmetric.

This course delivers the decomposition that has none of these defects: the **Singular Value Decomposition**,

$$A = U\Sigma V^T,$$

valid for **every** matrix A , square or rectangular, symmetric or not. Here U and V are orthogonal and Σ is a non-negative diagonal stretch. It is the most general, most stable, and arguably most useful matrix factorization in existence. From it we read off, in one stroke, the rank, the four fundamental subspaces, the matrix norms, the best low-rank approximation, the pseudo-inverse, the least-squares solution, and the principal components of a dataset.

The SVD is where the three previous courses converge. It is built **from** the spectral theorem (applied to the symmetric matrix $A^T A$); it **generalizes** eigendecomposition to all matrices; and it **is** the engine of principal component analysis, least squares, image compression, and numerical conditioning. Keep one picture in mind throughout: **the unit sphere, sent by any linear map to an ellipsoid** — the singular values are the lengths of its semi-axes, and everything in this course is a consequence of that single image.

Contents

Preface	2
1 The Singular Value Decomposition	4
1.1 Why eigendecomposition is not enough	4
1.2 Definition	4
1.3 Right singular vectors come from $A^T A$	5
1.4 Left singular vectors come from the right ones	5
1.5 The procedure, with a complete worked example	6
2 The geometry: rotate, stretch, rotate	7
2.1 Three motions	7
2.2 The unit sphere becomes an ellipsoid	7
3 The bridge to the spectral theorem	9
3.1 Symmetric positive semi-definite: SVD = spectral decomposition	9
3.2 The single difference: a negative eigenvalue becomes a sign	9
4 Matrix norms and the condition number	10
4.1 Two norms, both read off the singular values	10

4.2	The condition number	10
5	Low-rank approximation: the Eckart–Young theorem	11
5.1	Truncation	11
5.2	The theorem	12
5.3	The compression–quality tradeoff	12
6	The pseudo-inverse	13
6.1	Inverting what can be inverted	13
6.2	The Penrose conditions	13
6.3	Minimum-norm least squares	14
7	Least squares	14
7.1	An unsolvable system, solved as well as possible	14
7.2	The geometry: project b onto the column space	15
7.3	Conditioning of the least-squares solution	15
8	Principal Component Analysis	16
8.1	The problem and a first picture	16
8.2	The procedure	16
8.3	PCA is the SVD of the centered data	17
8.4	How much variance, and choosing k	17
9	When PCA is the wrong tool	18
9.1	Scale sensitivity	18
9.2	Linearity	18
9.3	Orthogonality	19
10	Numerical stability	19
10.1	Forming $A^T A$ squares the condition number	19
10.2	Why direct SVD algorithms win	20

1 The Singular Value Decomposition

We open with the object itself. The cleanest way to meet the SVD is to **want** it: to ask what would happen if we tried to diagonalize a matrix that is not square, and to discover that a small modification of the spectral theorem delivers exactly what we need.

1.1 Why eigendecomposition is not enough

Eigendecomposition asks: along which directions v does A act by pure scaling, $Av = \lambda v$? The question only makes sense when input and output live in the **same** space — that is, when A is square. For a 3×2 matrix the equation $Av = \lambda v$ is nonsense: the left side is in $\mathbb{R}^3$, the right in $\mathbb{R}^2$. Even for square matrices, the eigenvectors need not be orthogonal and need not span the space.

The SVD repairs both problems at once by allowing **two** orthonormal bases — one in the input space, one in the output space — instead of insisting on a single one.

Worked example — the SVD question. Instead of “which directions does A merely scale?”, ask: “is there an orthonormal basis $v_1, \dots, v_n$ of the input space that A sends to **orthogonal** output vectors?” If $Av_i = \sigma_i u_i$ with the u_i orthonormal and $\sigma_i \geq 0$, then in those two bases A is just a diagonal stretch — exactly the structure we are after. The SVD says such bases **always** exist.

1.2 Definition

Definition 1 (Singular value decomposition) Let $A \in M_{m,n}(\mathbb{R})$ have rank r . A **singular value decomposition** of A is a factorization

$$A = U\Sigma V^T,$$

where

- $U \in M_m(\mathbb{R})$ is **orthogonal** ($U^T U = I_m$),
- $V \in M_n(\mathbb{R})$ is **orthogonal** ($V^T V = I_n$), and
- $\Sigma \in M_{m,n}(\mathbb{R})$ is **rectangular diagonal** ($\Sigma_{ij} = 0$ for $i \neq j$) with diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots = 0.$$

The σ_i are the **singular values**; the columns $u_1, \dots, u_m$ of U are the **left singular vectors**; the columns $v_1, \dots, v_n$ of V are the **right singular vectors**.

The factorization says everything geometric about A in three readable pieces. The most useful way to see this is the **outer-product (dyadic) form**: multiplying out $U\Sigma V^T$ and keeping only the non-zero singular values gives

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T.$$

Each term $\sigma_i u_i v_i^T$ is a rank-one matrix; A is a sum of r of them, weighted by the singular values, **largest first**. This ordering — biggest pieces of A at the front — is the entire reason the SVD compresses, denoises, and reduces dimension. Hold onto it.

Common pitfall The ordering $\sigma_1 \geq \sigma_2 \geq \dots$ is part of the definition, not an afterthought. Singular values are **always** listed in decreasing order, so that “the first k singular values”

unambiguously means “the k largest”. Reorder them and you must reorder the columns of U and V to match — the i -th column of each goes with σ_i .

1.3 Right singular vectors come from $A^T A$

How could one ever build such a decomposition? The trick is to remove the asymmetry. The matrix A is rectangular, but $A^T A$ is a **square, symmetric** $n \times n$ matrix — exactly the kind the spectral theorem handles.

Proposition 2 (\$A^T A\$ is symmetric positive semi-definite) For any $A \in M_{m,n}(\mathbb{R})$, the matrix $A^T A$ is symmetric and positive semi-definite. Its eigenvalues are therefore real and ≥ 0 .

Proof. Symmetry: $(A^T A)^T = A^T (A^T)^T = A^T A$. Positive semi-definiteness: for any $x \in \mathbb{R}^n$,

$$x^T (A^T A) x = (Ax)^T (Ax) = \|Ax\|^2 \geq 0.$$

By the eigenvalue classification of MAT31-1, a symmetric matrix with $x^T M x \geq 0$ for all x has all eigenvalues ≥ 0 . $\square$

This is the hinge of the whole construction. Apply the spectral theorem (MAT31-1) to $A^T A$: there is an **orthonormal** basis $v_1, \dots, v_n$ of eigenvectors, with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$. These eigenvectors will be the right singular vectors, and the singular values will be the square roots of the eigenvalues.

Definition 3 (Singular values) The **singular values** of A are $\sigma_i = \sqrt{\lambda_i}$, where the λ_i are the eigenvalues of $A^T A$ in decreasing order. There are exactly $r = \text{rank}(A)$ non-zero ones.

The square root is the reason we form $A^T A$ rather than reading A directly: $A^T A$ converts the matrix into a symmetric object whose eigenvalues are the **squares** of the quantities we want. That squaring will return, as a warning, in the chapter on numerical stability — but as a **construction** it is exactly right.

1.4 Left singular vectors come from the right ones

Once the v_i are known, the left singular vectors are forced. The defining wish was $Av_i = \sigma_i u_i$, so we simply solve for u_i .

Proposition 4 (Left singular vectors) Let $v_1, \dots, v_n$ be orthonormal eigenvectors of $A^T A$ with eigenvalues $\lambda_i = \sigma_i^2$. For each i with $\sigma_i > 0$ define

$$u_i = \frac{1}{\sigma_i} Av_i.$$

Then $u_1, \dots, u_r$ are **orthonormal** in $\mathbb{R}^m$. Completing them to an orthonormal basis $u_1, \dots, u_m$ (any orthonormal basis of the remaining space will do) and assembling $U = (u_1 \mid \dots \mid u_m)$, $V = (v_1 \mid \dots \mid v_n)$, $\Sigma = \text{diag}(\sigma_1, \dots)$ gives $A = U \Sigma V^T$.

Proof. Orthonormality of the u_i is a direct computation using that the v_i are eigenvectors of $A^T A$:

$$\langle u_i, u_j \rangle = \frac{1}{\sigma_i \sigma_j} (Av_i)^T (Av_j) = \frac{1}{\sigma_i \sigma_j} v_i^T (A^T A) v_j = \frac{\lambda_j}{\sigma_i \sigma_j} v_i^T v_j = \frac{\sigma_j}{\sigma_i} \delta_{ij},$$

which is 1 if $i = j$ and 0 otherwise. Finally $Av_i = \sigma_i u_i$ holds for $\sigma_i > 0$ by definition, and for $\sigma_i = 0$ one checks $\|Av_i\|^2 = v_i^T A^T A v_i = \lambda_i = 0$, so $Av_i = 0 = \sigma_i u_i$ too. Thus $AV = U\Sigma$, and right-multiplying by $V^T = V^{-1}$ gives $A = U\Sigma V^T$. $\square$

The proof also proves existence: we never assumed an SVD, we **built** one.

Theorem 5 (Existence of the SVD) Every matrix $A \in M_{m,n}(\mathbb{R})$ admits a singular value decomposition.

1.5 The procedure, with a complete worked example

To compute an SVD of $A \in M_{m,n}(\mathbb{R})$:

1. Form $A^T A$; find its eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ and an **orthonormal** basis of eigenvectors $v_1, \dots, v_n$ (Gram–Schmidt inside any repeated eigenspace, MAT31-1).
2. Set $\sigma_i = \sqrt{\lambda_i}$ and $V = (v_1 \mid \dots \mid v_n)$.
3. For each $\sigma_i > 0$, set $u_i = Av_i/\sigma_i$. Complete $u_1, \dots, u_r$ to an orthonormal basis of $\mathbb{R}^m$ to obtain U .
4. Place the σ_i on the diagonal of an $m \times n$ matrix Σ . Then $A = U\Sigma V^T$.

Worked example — an SVD from start to finish. Let $A = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$.

Step 1 — the symmetric matrix.

$$A^T A = \begin{pmatrix} 2 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}.$$

Its eigenvalues solve $(5 - \lambda)^2 - 9 = 0$, giving $\lambda_1 = 8$, $\lambda_2 = 2$. The (orthonormal) eigenvectors are $v_1 = \frac{1}{\sqrt{2}}(1, 1)$ for $\lambda_1 = 8$ and $v_2 = \frac{1}{\sqrt{2}}(1, -1)$ for $\lambda_2 = 2$ — orthogonal, as the spectral theorem guarantees for distinct eigenvalues.

Step 2 — singular values. $\sigma_1 = \sqrt{8} = 2\sqrt{2}$, $\sigma_2 = \sqrt{2}$. So $\Sigma = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$.

Step 3 — left singular vectors.

$$Av_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \quad u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

$$Av_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

Both already orthonormal, so $U = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

Check.

$$U\Sigma V^T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix} = A. \quad \checkmark$$

Common pitfall The left singular vectors are **not** free to choose by eigen-analysis of $A^T A$ — they are determined by $u_i = Av_i/\sigma_i$. A frequent error is to separately diagonalize AA^T for the u_i and pair them with the v_i in the wrong sign or order; the signs then fail to match and $U\Sigma V^T \neq A$. Compute the u_i **from** the v_i . (Diagonalizing AA^T does give the u_i as a set — the same non-zero eigenvalues appear — but the pairing and signs must still come from Av_i .)

Remark The SVD is not quite unique. If a singular value is repeated, the corresponding singular vectors can be rotated within their eigenspace; and even for a simple singular value, the pair (u_i, v_i) may be replaced by $(-u_i, -v_i)$. What is unique is the **list of singular values** and the geometric action they encode.

Exercises

1. Compute an SVD of $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 2 \end{pmatrix}$ by forming $A^T A$. What are the singular values and the rank?
2. Find the singular values of $A = \begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$. Why are they 3 and 2, not 3 and -2 ?
3. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, compute $A^T A$, its eigenvalues, and hence the two singular values (leave them as square roots).
4. Show that A and A^T have the **same** non-zero singular values. (Hint: relate the eigenvalues of $A^T A$ and AA^T .)

2 The geometry: rotate, stretch, rotate

The algebra of the previous chapter hides a picture so vivid that, once seen, the whole course becomes intuitive. Reading $A = U\Sigma V^T$ right-to-left as it acts on a vector x shows that **every linear map does the same three things in sequence**.

2.1 Three motions

To compute $Ax = U\Sigma V^T x$, apply the factors in order:

1. V^T — an **orthogonal** map: a rotation (possibly with a reflection) of the input. It changes no lengths and no angles; it merely re-aligns the axes so that the special directions v_i become the coordinate axes.
2. Σ — a **non-negative diagonal stretch**: it scales the i -th coordinate by σ_i (and, if $m \neq n$, pads with zeros or drops coordinates). This is the only step that distorts shape.
3. U — another **orthogonal** map: a rotation/reflection of the result into its final position in the output space.

Rotate, stretch, rotate. Every real linear map factors as an orthogonal map, then an axis-aligned scaling by the singular values, then another orthogonal map. All of a matrix's shape-distortion lives in the middle factor Σ ; the two orthogonal factors only reorient.

2.2 The unit sphere becomes an ellipsoid

The cleanest consequence — and the image to carry through the entire course — is what A does to the unit sphere.

Proposition 6 (Image of the unit sphere) The image under A of the unit sphere $S = \{x \in \mathbb{R}^n : \|x\| = 1\}$ is an **ellipsoid** in $\mathbb{R}^m$ whose semi-axes have lengths equal to the singular values $\sigma_1 \geq \dots \geq \sigma_r$ and point along the left singular vectors $u_1, \dots, u_r$.

Proof. A point of S is $x = \sum_i c_i v_i$ with $\sum_i c_i^2 = 1$ (coordinates in the orthonormal basis of right singular vectors). Then $Ax = \sum_{i=1}^r \sigma_i c_i u_i$, so its coordinates $y_i = \sigma_i c_i$ along the orthonormal u_i satisfy

$$\sum_{i=1}^r \left(\frac{y_i}{\sigma_i} \right)^2 = \sum_{i=1}^r c_i^2 \leq 1,$$

the equation of an ellipsoid with semi-axis σ_i along u_i . □

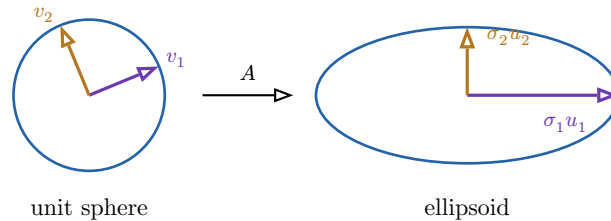


Figure 1: The defining picture of the SVD. The right singular vectors v_i are the orthonormal input directions that A sends to the orthogonal output directions u_i , stretched by the singular values σ_i . The unit sphere becomes an ellipsoid whose longest semi-axis is σ_1 (the most A can stretch any unit vector) and whose shortest is σ_r .

Worked example — the ellipse of our worked matrix. For $A = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$ we found $\sigma_1 = 2\sqrt{2} \approx 2.83$ along $u_1 = (1, 0)$ and $\sigma_2 = \sqrt{2} \approx 1.41$ along $u_2 = (0, -1)$. So the unit circle is mapped to an ellipse with a horizontal semi-axis of length $2\sqrt{2}$ and a vertical semi-axis of length $\sqrt{2}$. The most any unit vector is stretched is $2\sqrt{2}$ (achieved at $x = v_1$), the least is $\sqrt{2}$ (at $x = v_2$).

Remark This picture immediately interprets the largest and smallest singular values: $\sigma_1 = \max_{\|x\|=1} \|Ax\|$ is the **maximum gain** of A , and σ_r is the smallest gain over directions A does not kill. The ratio σ_1/σ_r — how much more A stretches its best direction than its worst — is the **condition number** of a later chapter, here made visible as the eccentricity of the ellipsoid.

Exercises

5. Sketch the image of the unit circle under $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$. What are the semi-axes and their directions?
6. A matrix A has singular values 5 and 0. What shape is the image of the unit circle? What does $\sigma_2 = 0$ say about A ?
7. Explain why an orthogonal matrix Q sends the unit sphere to itself, and deduce all its singular values.
8. Using the rotate–stretch–rotate reading, argue that $\det A = \pm \prod_i \sigma_i$ for a square A . Where does the sign come from?

3 The bridge to the spectral theorem

The SVD generalizes the spectral theorem, and it is worth seeing exactly **how** — where the two decompositions agree, and where the SVD’s extra generality shows up as the single difference of a sign.

3.1 Symmetric positive semi-definite: SVD = spectral decomposition

Proposition 7 (SVD of a symmetric PSD matrix) If A is symmetric positive semi-definite, its spectral decomposition $A = PDP^T$ (with P orthogonal and D diagonal with entries ≥ 0) is **already** a singular value decomposition: the singular values are the eigenvalues, and the left and right singular vectors coincide with the eigenvectors, $U = V = P$.

Proof. Here $A^T A = A^2 = PD^2 P^T$, so the eigenvalues of $A^T A$ are λ_i^2 and the singular values are $\sigma_i = \sqrt{\lambda_i^2} = |\lambda_i| = \lambda_i$ (using $\lambda_i \geq 0$). The eigenvectors of $A^T A$ are the columns of P , so $V = P$, and $u_i = Av_i/\sigma_i = \lambda_i v_i/\lambda_i = v_i$, giving $U = P$ as well. $\square$

So for the nicest matrices the SVD adds nothing new — it **is** the spectral theorem. The interest is in what happens when a symmetric matrix has a negative eigenvalue.

3.2 The single difference: a negative eigenvalue becomes a sign

Worked example — where eigenvalue and singular value part ways. Let $A = \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$, symmetric with eigenvalues -3 and 2 . Its singular values are $|-3| = 3$ and $|2| = 2$ — **positive**, as all singular values must be. The negative sign of the eigenvalue -3 has not vanished; it has moved into a singular vector. With $v_1 = (1, 0)$, $u_1 = Av_1/\sigma_1 = (-3, 0)/3 = (-1, 0) = -v_1$. The eigendecomposition records the reflection as a **negative scaling**; the SVD records it as a **positive stretch composed with a flip** (the flip absorbed into $u_1 = -v_1$).

The relationship in one line. For a symmetric matrix, singular values are the **absolute values** of the eigenvalues, and a negative eigenvalue’s sign is absorbed by setting $u_i = -v_i$. Hence the SVD coincides with the eigendecomposition **exactly when A is positive semi-definite** — precisely when there are no signs to absorb.

Common pitfall Do not conflate eigenvalues with singular values for non-PSD matrices. A symmetric matrix with eigenvalues $-3, 2$ has singular values $3, 2$; the largest singular value is 3 even though the largest eigenvalue is 2 . And for **non-symmetric** matrices the eigenvalues (when real) bear no simple relation to the singular values at all — e.g. $\begin{pmatrix} 1 & 100 \\ 0 & 1 \end{pmatrix}$ has both eigenvalues equal to 1 but singular values far from 1 .

Exercises

9. Give the singular values of $\begin{pmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ and the relation between each u_i and v_i .
10. For a symmetric matrix $A = PDP^T$ with eigenvalues $2, -2$, write an SVD explicitly and identify where the sign goes.
11. Prove that a symmetric matrix is positive semi-definite if and only if its singular values equal its eigenvalues.

4 Matrix norms and the condition number

The singular values are not only a geometric portrait of A ; they are also the right way to assign A a **size**. Three numbers built from the singular values — the Frobenius norm, the spectral norm, and the condition number — measure how big a matrix is, how much it can stretch, and how dangerous it is to invert.

4.1 Two norms, both read off the singular values

Definition 8 (Frobenius and spectral norms) For $A \in M_{m,n}(\mathbb{R})$ with singular values $\sigma_1 \geq \dots \geq \sigma_r$:

- the **Frobenius norm** is

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\sum_{i=1}^r \sigma_i^2};$$

- the **spectral norm** (or operator 2-norm) is

$$\|A\|_2 = \max_{\|x\|=1} \|Ax\| = \sigma_1.$$

The Frobenius norm treats the matrix as one long vector of its mn entries and takes the ordinary Euclidean length — this is the Frobenius inner product of MAT31-1. That it also equals $\sqrt{\sum \sigma_i^2}$ is the content of the next proposition: orthogonal factors U, V do not change the Euclidean length of the entries, so all the “size” sits in Σ .

Proposition 9 (Frobenius norm via singular values) $\|A\|_F^2 = \sum_{i=1}^r \sigma_i^2$.

Proof. The Frobenius norm satisfies $\|M\|_F^2 = \text{tr}(M^T M)$, and it is **unitarily invariant**: $\|QM\|_F = \|M\|_F = \|MQ\|_F$ for orthogonal Q , because $\text{tr}((QM)^T QM) = \text{tr}(M^T Q^T QM) = \text{tr}(M^T M)$. Hence $\|A\|_F = \|U\Sigma V^T\|_F = \|\Sigma\|_F = \sqrt{\sum_i \sigma_i^2}$. $\square$

The spectral norm is the **largest stretch** of the ellipsoid picture: by construction $\|Ax\| \leq \sigma_1 \|x\|$ with equality at $x = v_1$, so σ_1 is the maximum gain of A .

Worked example — the two norms of our matrix. For $A = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$, $\sigma_1 = 2\sqrt{2}$, $\sigma_2 = \sqrt{2}$. Then

$$\|A\|_2 = \sigma_1 = 2\sqrt{2} \approx 2.83, \quad \|A\|_F = \sqrt{8+2} = \sqrt{10} \approx 3.16.$$

Cross-check the Frobenius norm directly from the entries: $\sqrt{2^2 + 2^2 + 1^2 + 1^2} = \sqrt{10}$. $\checkmark$
Always $\|A\|_2 \leq \|A\|_F \leq \sqrt{r} \|A\|_2$, since one term dominates the sum but no term exceeds σ_1^2 .

4.2 The condition number

Definition 10 (Condition number) For a matrix A of full column rank with singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$, the **condition number** (in the spectral norm) is

$$\kappa(A) = \frac{\sigma_1}{\sigma_r}.$$

For a square invertible A this is $\sigma_1/\sigma_n = \|A\|_2 \|A^{-1}\|_2$.

The condition number is the eccentricity of the ellipsoid: how much more A stretches its most-stretched direction than its least-stretched one. It is always ≥ 1 , equals 1 exactly for (scalar multiples of) orthogonal matrices, and becomes large when A nearly collapses some direction.

Proposition 11 (Why the condition number measures sensitivity) Suppose A is invertible and $Ax = b$. If b is perturbed to $b + \delta b$, the solution changes to $x + \delta x$ with

$$\frac{\|\delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\delta b\|}{\|b\|}.$$

A relative error in the data can be amplified by up to a factor $\kappa(A)$ in the answer.

Proof. From $A\delta x = \delta b$ we get $\|\delta x\| \leq \|A^{-1}\| \|\delta b\|$, and from $b = Ax$, $\|b\| \leq \|A\| \|x\|$. Multiply: $\|\delta x\| \|b\| \leq \|A^{-1}\| \|A\| \|\delta b\| \|x\|$, which rearranges to the claim with $\kappa(A) = \|A\| \|A^{-1}\|$. $\square$

Common pitfall A large condition number is a property of the **problem**, not of the algorithm: no amount of cleverness can recover information that A has nearly destroyed by squashing a direction. When σ_r is tiny relative to σ_1 , the system is **ill-conditioned** and its solution should be distrusted — even if a computer returns one to fifteen digits. This single number is why the SVD, which exposes σ_r directly, is the diagnostic tool of numerical linear algebra.

Exercises

12. Compute $\|A\|_F$, $\|A\|_2$, and $\kappa(A)$ for $A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$.
13. Show $\kappa(A) \geq 1$ always, and that $\kappa(Q) = 1$ for any orthogonal Q .
14. A matrix has singular values 100, 10, 0.01. Is it well- or ill-conditioned? Give $\kappa(A)$.
15. Prove that $\|A\|_F = \|A\|_2$ if and only if A has rank at most 1.

5 Low-rank approximation: the Eckart–Young theorem

We now harvest the single most exploited consequence of the singular-value ordering. Truncating the dyadic sum $A = \sum \sigma_i u_i v_i^T$ after the k largest terms gives the **best possible** approximation of A by a matrix of rank k . This one theorem underlies image compression, recommender systems, noise removal, and the dimensionality reduction of the next chapter.

5.1 Truncation

Definition 12 (Truncated SVD) For $k \leq r$, the **rank- k truncation** of $A = \sum_{i=1}^r \sigma_i u_i v_i^T$ is

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T = U \Sigma_k V^T,$$

where Σ_k keeps only the k largest singular values and zeros the rest. The matrix A_k has rank exactly k .

Because the terms are ordered by decreasing σ_i , A_k keeps the “biggest pieces” of A and discards the smallest. The hope is that the discarded pieces are small — and the next theorem says that keeping the top k is not merely a reasonable heuristic but **provably optimal**.

5.2 The theorem

Theorem 13 (Eckart–Young) Among all matrices B of rank at most k , the truncated SVD A_k is a best approximation of A in both the spectral and the Frobenius norm:

$$\min_{\text{rank}(B) \leq k} \|A - B\|_2 = \|A - A_k\|_2 = \sigma_{k+1},$$

$$\min_{\text{rank}(B) \leq k} \|A - B\|_F = \|A - A_k\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2}.$$

The error is **exactly the discarded singular values**: in the spectral norm, the first one you threw away, σ_{k+1} ; in the Frobenius norm, the root-sum-square of all of them. You can therefore read the quality of a rank- k approximation **before computing it**, straight from the singular value list.

Worked example — the best rank-1 approximation. For $A = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$ with $\sigma_1 = 2\sqrt{2}$, $u_1 = (1, 0)$, $v_1 = \frac{1}{\sqrt{2}}(1, 1)$, the best rank-1 approximation is

$$A_1 = \sigma_1 u_1 v_1^T = 2\sqrt{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \frac{1}{\sqrt{2}}(1, 1) = \begin{pmatrix} 2 & 2 \\ 0 & 0 \end{pmatrix}.$$

The error is $\|A - A_1\|_2 = \sigma_2 = \sqrt{2}$, and indeed $A - A_1 = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}$ has spectral norm $\sqrt{2}$. ✓ No other rank-1 matrix comes closer to A .

5.3 The compression–quality tradeoff

Storing a general $m \times n$ matrix costs mn numbers. Storing its rank- k truncation costs only the u_i , v_i , and σ_i :

$$k(m + n + 1) \text{ numbers.}$$

When $k \ll \min(m, n)$ this is a large saving. The decay of the singular values decides whether it is worth it: if they fall off fast, a small k captures almost all of A and compression is nearly lossless; if they decay slowly, no low-rank approximation is faithful.

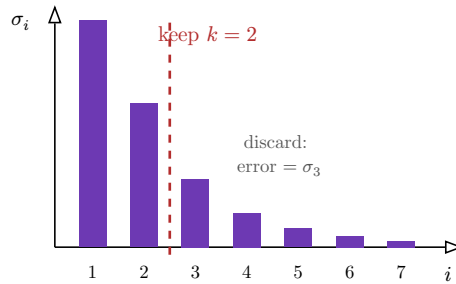


Figure 2: A **scree plot** of singular values, sorted in decreasing order. Rapidly decaying singular values mean a few rank-one terms carry almost all of A : truncating at $k = 2$ here discards only the small tail, and the spectral-norm error is exactly σ_3 . A good choice of k sits at the “elbow” where the values flatten out.

Common pitfall Eckart–Young is optimal only among **low-rank** approximations and only in the two norms stated. It does not say A_k is close to A in absolute terms — that depends

entirely on how fast the singular values decay. If they decay slowly (e.g. all roughly equal), then σ_{k+1} is not small and **no** rank- k matrix approximates A well; the data is genuinely high-dimensional. Always look at the singular value spectrum before trusting a low-rank model.

Exercises

16. Given singular values 10, 8, 1, 0.5, find the spectral- and Frobenius-norm errors of the best rank-2 approximation.

17. Compute the best rank-1 approximation of $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 2 \end{pmatrix}$ and its error.

18. For an $m \times n$ image with $m = n = 1000$, at what k does a rank- k approximation start to cost **more** storage than the original?

19. Explain, using Eckart–Young, why adding small random noise to a low-rank matrix and then truncating the SVD can **remove** much of the noise.

6 The pseudo-inverse

Not every matrix has an inverse — rectangular ones never do, and square singular ones do not either. The SVD nonetheless produces the closest possible substitute: the **Moore–Penrose pseudo-inverse**, which inverts A wherever inversion makes sense and does the most sensible thing where it does not.

6.1 Inverting what can be inverted

The idea is transparent in the SVD. Since $A = U\Sigma V^T$ with U, V orthogonal, inverting A should mean inverting each factor: V and U invert by transposing, and Σ inverts by reciprocating its non-zero entries. The only subtlety is the **zero** singular values — directions A destroys, which cannot be undone — and the pseudo-inverse simply leaves those as zero.

Definition 14 (Moore–Penrose pseudo-inverse) Let $A = U\Sigma V^T$. The **pseudo-inverse** of A is

$$A^+ = V\Sigma^+U^T,$$

where $\Sigma^+ \in M_{n,m}(\mathbb{R})$ is formed from Σ by transposing it and replacing each non-zero σ_i by $1/\sigma_i$ (leaving every zero in place).

Worked example — a pseudo-inverse. For $A = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$ (here actually invertible), with $U = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$, $V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$:

$$\Sigma^+ = \begin{pmatrix} \frac{1}{2\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad A^+ = V\Sigma^+U^T = \frac{1}{4}\begin{pmatrix} 1 & -2 \\ 1 & 2 \end{pmatrix}.$$

Since A is invertible, $A^+ = A^{-1}$ — one checks $AA^+ = I$. The pseudo-inverse **generalizes** the inverse, agreeing with it whenever the inverse exists.

6.2 The Penrose conditions

A remarkable fact is that this SVD recipe produces the **unique** matrix satisfying four simple algebraic identities — a definition that never mentions the SVD at all.

Proposition 15 (Penrose conditions) $X = A^+$ is the unique matrix satisfying

$$AXA = A, \quad XAX = X, \quad (AX)^T = AX, \quad (XA)^T = XA.$$

The first two conditions say A and X are “weak inverses” of each other; the last two say the products AX and XA are **symmetric** — they are in fact the orthogonal projections onto the column space of A and the row space of A respectively. This projection reading is what makes the pseudo-inverse solve least-squares problems, as we see next.

6.3 Minimum-norm least squares

Proposition 16 (What A^+ computes) For any b , the vector $x^* = A^+b$ is the least-squares solution of $Ax = b$ of **smallest norm**: it minimizes $\|Ax - b\|$, and among all minimizers it is the unique one with the smallest $\|x\|$.

When $Ax = b$ has no solution, A^+b gives the best approximate one; when it has **many** solutions, A^+b singles out the shortest. Both behaviours fall out of inverting only the non-zero singular values and zeroing the rest — the pseudo-inverse ignores precisely the directions where the data is silent.

Common pitfall A^+ is **not** $(A^T A)^{-1} A^T$ in general — that formula is valid only when A has full column rank (so $A^T A$ is invertible). The SVD definition $A^+ = V \Sigma^+ U^T$ always works, including for rank-deficient A where $A^T A$ is singular and the normal-equations formula breaks down.

Exercises

20. Verify the four Penrose conditions for $A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$ and $A^+ = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{pmatrix}$.
21. Compute A^+ for $A = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (a 2×1 matrix). What does A^+b compute geometrically?
22. Show that if A has full column rank, then $A^+ = (A^T A)^{-1} A^T$.
23. Prove that $(A^+)^+ = A$ using the SVD.

7 Least squares

The pseudo-inverse points straight at one of the most important problems in applied mathematics: solving a system that has **no** solution. Over-determined systems — more equations than unknowns — are the rule whenever we fit a model to noisy data, and the least-squares method is how we extract the best answer anyway.

7.1 An unsolvable system, solved as well as possible

Worked example — fitting a line. Suppose three data points $(1, 1)$, $(2, 2)$, $(3, 2)$ should lie on a line $y = a + bx$. Demanding exact fit gives three equations in two unknowns:

$$a + b = 1, \quad a + 2b = 2, \quad a + 3b = 2,$$

i.e. $A \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$ with $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$. No line passes through all three points, so this system is

inconsistent. We cannot make the residual zero — so we make it as **small** as possible instead.

Definition 17 (Least-squares problem) Given $A \in M_{m,n}(\mathbb{R})$ (typically $m > n$) and $b \in \mathbb{R}^m$, the **least-squares problem** is to find x minimizing the residual norm $\|Ax - b\|$. The vector $r = b - Ax$ is the **residual**.

7.2 The geometry: project b onto the column space

Minimizing $\|Ax - b\|$ means finding the point Ax in the column space of A that is **closest** to b — exactly the best-approximation problem of MAT31-1. The answer there was the orthogonal projection, characterized by the residual being perpendicular to the subspace.

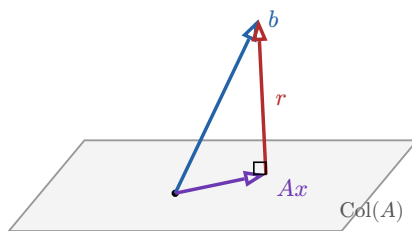


Figure 3: Least squares is orthogonal projection. The achievable outputs Ax fill the column space of A (grey plane). The closest one to the target b is its orthogonal projection; the residual $r = b - Ax$ is **perpendicular** to the column space, which is the condition $A^T r = 0$.

Theorem 18 (Normal equations and residual orthogonality) A vector x minimizes $\|Ax - b\|$ if and only if it satisfies the **normal equations**

$$A^T Ax = A^T b, \quad \text{equivalently} \quad A^T r = 0 \quad \text{with } r = b - Ax.$$

That is, the residual is orthogonal to the column space of A . If A has full column rank, the minimizer is unique and

$$x = (A^T A)^{-1} A^T b = A^+ b.$$

Proof. The closest point of the subspace $\text{Col}(A)$ to b is its orthogonal projection (best-approximation theorem, MAT31-1), characterized by $b - Ax \perp \text{Col}(A)$. Orthogonality to every column of A is the single equation $A^T(b - Ax) = 0$, i.e. $A^T Ax = A^T b$. When A has full column rank $A^T A$ is invertible, giving the stated unique solution; one checks it coincides with $A^+ b$. $\square$

Worked example — the line, fitted. For the data above, $A^T A = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}$ and $A^T b = \begin{pmatrix} 5 \\ 11 \end{pmatrix}$. Solving $\begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \end{pmatrix}$ gives $b = \frac{1}{2}$, $a = \frac{2}{3}$, so the best-fit line is $y = \frac{2}{3} + \frac{1}{2}x$. The residual at the three points is $r = (1, 2, 2) - (\frac{7}{6}, \frac{5}{3}, \frac{13}{6}) = (-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6})$, and indeed $A^T r = 0$. $\checkmark$

7.3 Conditioning of the least-squares solution

How much should we trust x ? Its reliability is governed by the condition number $\kappa(A)$: the more ill-conditioned A , the more a small perturbation of the data b swings the solution. This is the same σ_1/σ_r as before, and it is the reason the final chapter warns against forming $A^T A$ carelessly.

Common pitfall The normal equations are mathematically correct but numerically treacherous. They require forming $A^T A$, which — as the next chapter proves — **squares** the condition number. For well-conditioned problems this is harmless; for ill-conditioned ones

it can destroy half the available precision. Production least-squares solvers therefore avoid $A^T A$ and work from a direct factorization of A (QR or SVD).

Exercises

24. Set up and solve the normal equations to fit $y = a + bx$ to $(0, 1)$, $(1, 0)$, $(2, 2)$.
25. Verify, for your solution above, that the residual satisfies $A^T r = 0$.
26. Explain why the least-squares solution is unique exactly when A has full column rank, in terms of the null space of A .
27. Show that the minimal residual norm equals $\|(I - AA^+)b\|$, the distance from b to the column space.

8 Principal Component Analysis

We arrive at the application that makes the SVD indispensable to data science. **Principal Component Analysis** finds the directions of greatest variation in a dataset — the axes along which the data actually spreads — and lets us describe high-dimensional data with a handful of coordinates. It is the spectral theorem and the SVD, applied to a covariance matrix.

8.1 The problem and a first picture

Worked example — a cloud of points. Imagine a cloud of two-dimensional data points that, plotted, forms a long thin diagonal streak. Neither the x -axis nor the y -axis is special — the **spread** of the data lies along a tilted direction. PCA finds that direction (the long axis of the streak) as the **first principal component**, and the perpendicular direction (the thin width) as the second. Projecting onto just the first axis loses little, because the data barely varies in the second — a two-dimensional dataset compressed to one dimension with small loss.

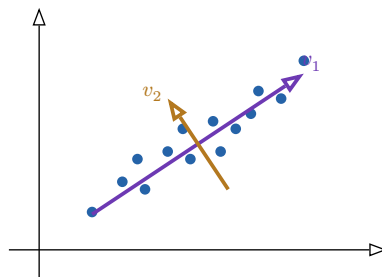


Figure 4: Principal components of a centered point cloud. The first principal component v_1 points along the direction of **greatest variance** (the long axis of the cloud); v_2 is orthogonal to it and captures the remaining spread. Projecting onto v_1 alone retains most of the variation, because the spread along v_2 is small.

8.2 The procedure

PCA of a data matrix $X \in M_{m,n}(\mathbb{R})$ (m observations in rows, n features in columns):

1. **Center.** Subtract from each column its mean, giving $\tilde{X}$ with zero-mean columns.
2. **Covariance.** Form the symmetric positive semi-definite matrix

$$C = \frac{1}{m-1} \tilde{X}^T \tilde{X} \in M_n(\mathbb{R}).$$

3. **Diagonalize.** By the spectral theorem, $C = V\Lambda V^T$ with V orthogonal and $\Lambda = \text{diag}(\lambda_1 \geq \dots \geq \lambda_n \geq 0)$. The eigenvectors $v_1, \dots, v_n$ are the **principal components**; λ_i is the variance captured by v_i .
4. **Reduce.** Keep the top k components; the reduced coordinates of an observation are its projections onto $v_1, \dots, v_k$.

Centering is not optional cosmetic preprocessing: the covariance matrix measures spread **about the mean**, so the data must be shifted to mean zero first. Skip it and the leading “principal component” merely points at the centroid, capturing the data’s **location** rather than its **variation**.

8.3 PCA is the SVD of the centered data

Forming C and diagonalizing it repeats, needlessly, the construction of the SVD — recall C is built exactly like the $A^T A$ of Chapter 1. The SVD of $\tilde{X}$ delivers the principal components directly.

Proposition 19 (PCA via the SVD) Let $\tilde{X} = U\Sigma V^T$ be an SVD of the centered data. Then

$$C = \frac{1}{m-1} \tilde{X}^T \tilde{X} = \frac{1}{m-1} V \Sigma^T \Sigma V^T,$$

so the principal components are the **right singular vectors** v_i of $\tilde{X}$, and the variances are $\lambda_i = \sigma_i^2 / (m-1)$.

This is why practitioners never form the covariance matrix: the SVD of $\tilde{X}$ gives the same v_i while avoiding the numerical hazard of $\tilde{X}^T \tilde{X}$ (Chapter 10). The principal components, the variances, and the reduced coordinates all read off one SVD.

8.4 How much variance, and choosing k

Definition 20 (Explained variance ratio) The **explained variance ratio** of the i -th principal component is

$$\frac{\lambda_i}{\sum_{j=1}^n \lambda_j} = \frac{\sigma_i^2}{\sum_{j=1}^n \sigma_j^2},$$

the fraction of the total variance it accounts for.

To choose the number of components k , pick the smallest k whose **cumulative** explained variance ratio

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^n \lambda_j}$$

reaches a target such as 0.90 or 0.95 — i.e. retain enough components to keep, say, 95% of the variance. The scree plot of Chapter 5 is exactly this picture: look for the elbow where the explained variance stops growing appreciably.

Worked example — reading explained variance. Suppose a 4-feature dataset has variances $\lambda = (6.0, 2.5, 1.0, 0.5)$, summing to 10. The first component explains 60%, the first two 85%, the first three 95%. To retain 95% of the variance one keeps $k = 3$ components, reducing four features to three with a 5% loss of variance.

Common pitfall Explained variance measures **variance retained**, not **information retained** for your task. A low-variance direction can still be the one that separates two classes; PCA, being unsupervised, will happily discard it. High cumulative explained variance justifies the compression numerically, but never confirms that the discarded directions were irrelevant to the downstream problem.

Exercises

28. A dataset has singular values (of the centered matrix) 4, 3, 1. Give the explained variance ratio of each component and the cumulative ratio for $k = 2$.
29. Explain why C is always symmetric positive semi-definite, so that its eigenvalues (the variances) are real and non-negative.
30. Why does PCA use the right singular vectors of $\tilde{X}$ rather than the left ones? What do the left singular vectors represent?
31. You forget to center the data. Describe concretely what the first “principal component” then tends to capture.

9 When PCA is the wrong tool

PCA is powerful precisely because it commits to three strong assumptions — and it fails, sometimes silently, when they do not hold. A competent analyst checks these before trusting the output. Each limitation is a direct consequence of **how** PCA is built.

9.1 Scale sensitivity

PCA acts on the covariance matrix, which is built from the raw feature values, so it depends on the **units** of each feature. A feature measured in grams has a thousand times the variance of the same feature in kilograms, and PCA will treat the larger-variance feature as more important purely because of the unit choice.

Worked example — units hijack the components. A dataset has “height in metres” (values around 1.7, variance ≈ 0.01) and “income in euros” (values around 40000, variance $\approx 10^8$). The income variance dwarfs the height variance, so the first principal component points almost entirely along income — not because income is more informative, but because euros are small units. The remedy is to **standardize** each feature to unit variance (divide by its standard deviation) before running PCA, putting all features on a common footing.

9.2 Linearity

Principal components are straight lines (directions); PCA can only capture **linear** structure. Data lying on a curved manifold — points on a spiral, on a circle, on an S-curve — has its true low-dimensional structure invisible to PCA.

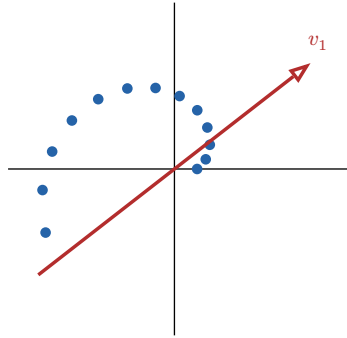


Figure 5: A case where PCA fails. The points lie on a one-dimensional **curved** manifold (a spiral), so a single number — arc length along the curve — describes them. But PCA can only offer a straight first component v_1 (red), which collapses distant parts of the spiral together and captures the structure poorly. Linear methods cannot unroll a nonlinear manifold.

9.3 Orthogonality

PCA forces its components to be mutually **orthogonal** (the spectral theorem guarantees orthogonal eigenvectors). If the genuine underlying factors are correlated — not at right angles — PCA cannot represent them faithfully; it will return orthogonal mixtures of the true factors instead. Techniques that drop the orthogonality constraint (independent component analysis, non-negative matrix factorization) exist precisely for this case.

Before applying PCA, check:

- **Scale** — are features on comparable units, or should they be standardized first?
- **Linearity** — does the data plausibly lie near a linear subspace, or on a curved manifold?
- **Orthogonality** — is it reasonable for the dominant factors to be mutually perpendicular?

If any answer is “no”, PCA’s output may mislead even while its explained-variance numbers look healthy.

Exercises

32. Two features have variances 0.01 and 10^6 in their given units. Explain what the first principal component captures, and how standardization changes it.
33. Give an example of a dataset (in words) whose intrinsic dimension is 1 but for which PCA needs 2 or more components to reconstruct well.
34. Why does the orthogonality of principal components follow from the spectral theorem applied to the covariance matrix?

10 Numerical stability

We close by paying a debt incurred in Chapter 1. To **build** the SVD we formed $A^T A$ and took square roots of its eigenvalues — a clean construction on paper. On a finite-precision computer, forming $A^T A$ is dangerous, and understanding why is the final piece of the course.

10.1 Forming $A^T A$ squares the condition number

Proposition 21 (Conditioning of $A^T A$) For any matrix A , the singular values of $A^T A$ are the squares σ_i^2 of the singular values of A . Consequently, for full-column-rank A ,

$$\kappa(A^T A) = \frac{\sigma_1^2}{\sigma_r^2} = \kappa(A)^2.$$

Proof. If $A = U\Sigma V^T$ then $A^T A = V\Sigma^T \Sigma V^T = V(\Sigma^T \Sigma)V^T$, a spectral decomposition with eigenvalues σ_i^2 . These are also its singular values (it is symmetric PSD), so $\kappa(A^T A) = \sigma_1^2/\sigma_r^2 = (\sigma_1/\sigma_r)^2 = \kappa(A)^2$. $\square$

The consequence is sharp. If A has condition number 10^6 — large but not catastrophic — then $A^T A$ has condition number 10^{12} . In IEEE double precision (about 16 significant digits) that is enough to obliterate roughly half the available precision. The normal equations of Chapter 8 and the covariance matrix of Chapter 9 both form exactly this product, and both inherit exactly this hazard.

Worked example — lost digits. Let $A = \begin{pmatrix} 1 & 1 \\ \varepsilon & 0 \\ 0 & \varepsilon \end{pmatrix}$ with $\varepsilon = 10^{-8}$. Then

$$A^T A = \begin{pmatrix} 1 + \varepsilon^2 & 1 \\ 1 & 1 + \varepsilon^2 \end{pmatrix}.$$

On a machine that rounds $1 + \varepsilon^2 = 1 + 10^{-16}$ back to 1, the computed $A^T A$ becomes $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ — **singular**, with the information in ε entirely lost. Working from A directly, that information survives: the smallest singular value of A is about $\varepsilon\sqrt{2}$, small but representable.

10.2 Why direct SVD algorithms win

The lesson. Whenever the mathematics tempts you to form $A^T A$ — normal equations, covariance matrix, “just diagonalize $A^T A$ to get the SVD” — a finite-precision computer pays by squaring the condition number. **Direct SVD algorithms** compute the singular values of A **without ever forming $A^T A$** , operating at condition number $\kappa(A)$ rather than $\kappa(A)^2$. For ill-conditioned problems this is the difference between a trustworthy answer and numerical garbage.

This is the practical reason the SVD, and not the eigendecomposition of $A^T A$, is the foundation of real numerical linear algebra. The construction via $A^T A$ in Chapter 1 is the right way to **understand** the SVD and to compute it **by hand** on small exact matrices; direct SVD algorithms (based on orthogonal transformations of A itself, beyond our scope) are the right way to **compute** it on a machine. The mathematics and the numerics agree on the answer; they disagree only on the road to it, and the condition number is the toll.

Common pitfall “Numerically unstable” does not mean “wrong formula”. The normal equations give the exact least-squares solution in exact arithmetic, and forming $A^T A$ gives the exact SVD in exact arithmetic. Instability is about **amplification of rounding error**, and it bites only for ill-conditioned data. For well-conditioned problems ($\kappa(A)$ modest) the simple $A^T A$ route is perfectly fine — know your condition number before deciding.

Exercises

- 35.** If $\kappa(A) = 10^4$ and your machine carries 16 digits, roughly how many digits of accuracy do you risk losing by solving via the normal equations?
- 36.** Prove that $\kappa(A^T A) = \kappa(A)^2$ for any full-column-rank A directly from the singular values.
- 37.** Explain why orthogonal transformations are the building block of stable numerical algorithms, recalling that orthogonal matrices have condition number 1.
- 38.** A colleague forms the covariance matrix $C = \tilde{X}^T \tilde{X} / (m - 1)$ and is surprised that small variances come out as exactly zero. Diagnose the issue using this chapter.